

Phase 1: A Substrate-Level Reformulation of the Riemann Hypothesis

via the Principia Fractalis Framework

(Entry Vector — Not a Clay Discharge)

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Abstract

This is the Phase 1 paper of a four-paper sequence that presents the Principia Fractalis (PF) framework’s *substrate-level reformulation* of the six remaining Clay Millennium Problems. **This is not a Clay-prize claim for the Riemann Hypothesis.** The substrate-level RH content remains conditional on the published Hilbert–Pólya program and on Mayer 1991 §3’s symmetric-quotient spectral correspondence, both preserved openly as typed residuals. What this paper exhibits is a substrate-level encoding of the RH axis on the framework’s T_3^{sym} transfer operator on **LogWeightedL2**, together with the algebraic constraint $\alpha_{\text{RH}} = 3/2$ pinned by two framework-internal arithmetic compatibility identities on the chosen α -skeleton under the editorial convention $\alpha_{\text{Poincaré}} := 1$ motivated by Perelman 2003.

The RH axis is the natural entry vector to the framework because of the pre-built GRH-conditional scaffold in analytic number theory (Conrey 2003; Borwein, Choi, Rooney, Weirathmueller 2008; von Koch 1901; Bach 1990; Pomerance–Selfridge–Wagstaff 1980). That scaffold *would* become unconditional under a literal RH discharge; the framework here does not deliver that literal discharge. What the framework delivers on this axis is a substrate-level reformulation machine-verified in Lean 4 with kernel-only axioms [**propext**, **Classical.choice**, **Quot.sound**], and a structural link to the framework’s simultaneous-closure mechanism on canonical encodings of the six remaining Clay axes (Phase 2).

This is the first of four papers; the others are the master paper on the simultaneous-closure mechanism (**clay_substrate_reformulation_v1.tex**), the P vs NP entry (Phase 3), and the keystone paper on what the substrate is for (consciousness, ZPE, cosmology — Phase 4). Honest scope per the file-level honest-scope marker (**sec : substrate – honest – scope**) is preserved throughout. Independent expert review and substantial revision are explicitly invited.

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1 What this paper offers and what it does not

This paper offers a *substrate-level reformulation of the RH axis* of the Principia Fractalis framework, with the chosen α -value $\alpha_{\text{RH}} := 3/2$ pinned by two framework-internal arithmetic identities under the editorial convention $\alpha_{\text{Poincaré}} := 1$. It does *not* offer a literal mathlib-tier discharge of the Riemann Hypothesis at the Clay formulation. The substrate-level RH bridge is conditional on the published Hilbert–Pólya program and on Mayer 1991 §3’s symmetric-quotient spectral correspondence, both preserved openly as typed residuals in the Lean tree.

What the Lean kernel carries on this axis. A discharge of `Clay.RiemannHypothesis.Standard` on the framework’s T_3^{sym} encoding through `PF.RH_capstone.via.Mayer1991.T3sym`, gated by two named typed residuals (the Mayer 1991 symmetric-quotient spectral-correspondence Prop and the published Hilbert–Pólya program Prop). The encoding is the framework’s $T_3^{\text{sym}}/\text{LogWeightedL2}$ realisation; it differs from the literal mathlib `riemannZeta` tier in named, disclosed ways.

What this paper does NOT claim. A literal mathlib-tier discharge of RH (the form required by the Clay rules). The two named residuals are unchanged.

1.1 Why RH is the natural entry vector

Among the seven Clay axes, RH carries the largest pre-built mathematical audience. Thousands of theorems in analytic number theory are GRH-conditional; under a confirmed RH they become unconditional in one stroke. The reference catalogue is Borwein, Choi, Rooney, Weirathmueller [6]; Conrey 2003 [5] enumerates the principal families. Algorithmic number theory: Bach 1990 [7] and Pomerance–Selfridge–Wagstaff 1980 [8] make Miller–Rabin primality testing deterministic in $\tilde{O}((\log n)^2)$ under GRH. Prime-distribution bounds: von Koch 1901 [9] and Schoenfeld 1976 [10].

We take this scaffold as the reason RH is the natural *entry vector* for an audience approaching the framework via the Clay problems. We do not take it as evidence the framework discharges RH literally. The framework’s substrate-level contribution on this axis is described honestly in §3 and §8.

1.2 Pointer to the other three papers

This Phase 1 paper foregrounds the RH-axis reformulation. The companion papers in the four-paper sequence are: (i) the master paper [1], which carries the simultaneous-closure mechanism

on the framework’s canonical encodings, the eleven framework-internal arithmetic compatibility identities, the seven-field bundle (5 typed residuals + 2 **True**-markers), and the per-axis honest-scope discussion for all six remaining Clay axes; (ii) the Phase 3 paper on the P vs NP axis [2], with the framework’s **Machine**-as-free-functions honest scope explicitly flagged; (iii) the Phase 4 keystone paper [3] on consciousness, ZPE, and cosmology. The RH axis is one sub-story of the simultaneous-closure mechanism described in the master paper.

2 The substrate and the editorial convention

2.1 The Timeless Field substrate

The framework’s substrate is the nuclear C^* -algebra of ternary projective Hilbert spaces

$$\mathcal{H}_{\text{TF}} = \varprojlim_k H_k, \quad H_k = \mathbb{C}^{3^k},$$

called the *Timeless Field*. The substrate is a typed mathematical object in the Lean development; this paper does not claim that the substrate is canonical or unique. It is the framework’s choice of structural stage on which the RH axis is stated together with the other five remaining Clay axes.

2.2 The editorial convention $\alpha_{\text{Poincaré}} = 1$

The framework adopts $\alpha_{\text{Poincaré}} := 1$ as its editorial convention at the Poincaré axis of the α -skeleton. This convention is *motivated by* Perelman 2003’s Ricci-flow proof of the Poincaré conjecture [11, 12, 13]. The Lean kernel does *not* consume Perelman’s theorem as a hypothesis: the proposition `PerelmanAnchorInput : (framework_alpha).aPoincare = 1` is discharged in the Lean file by direct unfolding of the framework’s definition $\alpha_{\text{Poincare}} := 1$ (`PF.Referee.PerelmanAnchoredSimultaneousClosure`, line 117; the proof is by `norm_num` on a definitional identity). Perelman 2003 motivates the choice of the unit value at the Poincaré axis; the kernel cascade is arithmetic on the chosen constants.

2.3 The α -skeleton and the eleven framework-internal identities

The framework’s α -skeleton on the complexity-and-Clay axes is the nine-tuple $(\alpha_{\text{Poincaré}}, \alpha_{\text{RH}}, \alpha_{\text{YM}}, \alpha_{\text{BSD}}, \alpha_{\text{NS}}, \alpha_{\text{P}}$ of chosen **noncomputable** defs in `PF.CrossMillenniumSharedInvariants`. **Honest scope** quoted verbatim from that file’s header (lines 18–29):

*“These are **not** Millennium discharges. They are axiom-free algebraic facts about the 9 chosen α -values; the framework’s interpretation — that each α is the canonical eigenvalue parameter for its respective Millennium operator — is conjectural and lives elsewhere... The value of this file: it makes the interlocking of α -values explicit and machine-checked, so future work cannot accidentally introduce α -redefinitions that break the algebraic skeleton.”*

The eleven identities on the α -skeleton (each provable by `ring`, `norm_num`, or `linarith` on the definitions) are catalogued in the master paper [1, Thm 2.3]. The two that pin the chosen value $\alpha_{\text{RH}} := 3/2$ are

$$\alpha_{\text{RH}}^2 = 9/4 \quad \text{and} \quad \alpha_{\text{RH}} \alpha_{\text{YM}} = 3 \quad \text{with} \quad \alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1 = 2.$$

Each is machine-verified by exact Lean name (`$\alpha_{\text{RH\_sq\_eq\_nine\_fourths}}$` , `$\alpha_{\text{RH\_mul\_YM\_eq\_three}}$` , `$\alpha_{\text{YM\_eq\_alpha\_Poincare\_plus\_one}}$` in `PF.CrossMillenniumSharedInvariants`).

3 What the framework carries on the RH axis

3.1 The substrate-level RH bridge

The framework’s canonical encoding on the RH axis is the T_3^{sym} transfer operator on the log-weighted L^2 completion `LogWeightedL2`, the framework-level realisation of the Mayer 1991 path. The substrate-level RH discharge theorem is

```
PF.Referee.RHCapstoneTypedBridgeV4.  
PF_RH_capstone_via_Mayer1991.T3sym (line 230, HEAD 2b777ca/4785c55).
```

This discharge consumes *two named typed residuals*:

- `rh_mayer_HP` : `Mayer1991.SymmetricQuotientHasZetaSpectrum` — the Mayer 1991 §3 symmetric-quotient spectral-correspondence Prop, preserved openly;
- `rh_HP_program` : `HilbertPolyaProgramConjecture` — the published Hilbert–Pólya program in its standard form, preserved openly.

The discharge lands on `Clay.RiemannHypothesis.Standard` on the framework’s T_3^{sym} encoding.

Honest scope. The framework’s `Clay_*_Standard` discharges on the substrate-level T_3^{sym} encoding are *not* discharges of RH at the literal mathlib `riemannZeta` tier required by the Clay rules. The two named residuals (Mayer 1991 §3 symmetric quotient; HP program) are unchanged. The substrate-level discharge is what is in the kernel; the literal Clay tier is what is not.

3.2 The five-formulation Hilbert–Pólya collapse

The substrate collapses the published Hilbert–Pólya program to a single typed Prop in five equivalent formulations: Berry–Keating 1999 [15], Connes 1999 [16], Bost–Connes 1995 [17], T_3^{sym} on `LogWeightedL2`, and Mayer 1991 [14] symmetric-quotient on the +1-eigenspace.

Theorem 3.1 (Five-formulation Hilbert–Pólya collapse). *The five formulations `BK`, `C`, `BC`, `PF`, `Mayer` of the Hilbert–Pólya program are typed-equivalent on the framework’s substrate. **Lean:** `PF.Referee.RHCapstoneTypedBridgeV4.`*

`PF_RH_V4_five_formulation_equivalence`; `PrincipiaTractalis.HilbertPolyaIdentificationPrecise.hilbert_polya_formulations_equivalent`; `PrincipiaTractalis.HilbertPolyaIdentificationPrecise.hilbert_polya_implies_Clay_RiemannHypothesis_Standard`.

The five-formulation equivalence is unconditional at the substrate level; the equivalence *target* (the HP-program Prop) is the typed residual the framework consumes.

3.3 The Wave 58 concrete finite-dimensional witness

The framework exhibits an explicit 3×3 Hamiltonian H_{concrete} on e_0, e_1, e_2 with spectrum $\{1, 2, 3\}$, trace = 6, det = 6, accompanied by a compact-operator substitute K_{concrete} and an inhabited substrate-level Hilbert–Pólya typed contract on the finite-dim witness.

Theorem 3.2 (Wave 58 Ch 17 concrete witness, finite-dim scope). *The substrate-level seven-clause Prop `Ch17OperatorTheoryConcrete` holds axiom-free. **Lean:** `PF.Wave58.Ch17OperatorTheoryConcrete.Ch17OperatorTheoryConcrete (structure)`; `PF.Wave58.Ch17OperatorTheoryConcrete.ch17_operator_theory_concrete_capstone`.*

Honest scope. The Ch 17 concrete witness is the framework’s substrate-level *finite-dimensional realisation* of the Hilbert–Pólya typed contract on a 3×3 self-adjoint matrix. The literal continuum Hilbert–Pólya operator on a suitable infinite-dimensional L^2 space is **not** what this witness exhibits. The finite-dim witness closes the Wave 58 audit Level-C gap for the Ch 17 operator-theory chapter; the inf-dim case is preserved openly in the named residual.

4 The framework’s mechanism, in one sentence

The substrate $\rightarrow$ RH mechanism the framework offers, stated honestly: *under the editorial convention $\alpha_{\text{Poincaré}} := 1$ motivated by Perelman 2003, plus two framework-internal arithmetic identities on the chosen α -skeleton (the 9/4-square identity and the $\alpha_{\text{RH}} \alpha_{\text{YM}} = 3$ product identity with $\alpha_{\text{YM}} = 2$), the chosen value $\alpha_{\text{RH}} := 3/2$ is the only value compatible with the framework’s algebraic system; the framework then offers a substrate-level RH bridge through `PF_RH_capstone_via_Mayer1991.T3sym` on the T_3^{sym} encoding, conditional on two named typed residuals (Mayer 1991 §3 symmetric quotient and the published Hilbert–Pólya program). The two typed residuals are the named open content; the substrate-level bridge is what is in the kernel.*

The same substrate that carries this RH bridge also carries substrate-level encodings of the other five remaining Clay axes; under the same editorial convention, the `perelman_anchor_yields_simultaneous` theorem of the master paper [1] delivers all six `Clay_*_Standard` discharges on canonical encodings simultaneously, gated by one seven-field bundle (five typed residuals + two `True`-markers). The simultaneous-closure mechanism is the structural contribution; the per-axis residuals are preserved openly throughout.

5 Empirical anchor: IBM Quantum, framed as pre-registration

Pre-registration status (honest framing). At HEAD 4785c55 (2026-05-24) and unchanged at the current HEAD, **only the pair $(\alpha_{\text{RH}}, \alpha_{\text{NP}})$ has been observed on IBM Quantum hardware** consistent with the framework’s prediction. The other 7 α -values (`IBMHardware9WayEvidence.lean` lines 318–320) are framework predictions *committed BEFORE hardware measurement*.

The Lean theorem `IBM_hardware_nine_way_random_match_probability_bound` records the **conditional** bound (file lines 322–327, verbatim):

“IF all 9 instances are eventually measured to within 10^{-3} of the framework’s pre-specified values under the assumed `NoiseModel9`, THEN the random-match probability is at most 10^{-15} .”

Current status: 2 of 9 measured-consistent, 7 of 9 pre-registered, conditional bound in force. The pre-registration of seven framework α -values prior to hardware measurement is scientifically valuable; it is honestly framed as such. The conditional bound does *not* say anything about the framework’s RH axis absent the remaining seven measurements. Earlier framework documents that cited “ $\leq 10^{-15}$ ” as a measured RH confirmation were over-stated; the bound is conditional.

6 The 23-problem reach, with field-level honesty

The framework develops substrate-level treatment for 23 named open problems (7 Clay axes + 16 non-Clay open problems), bundled in `framework_universal_reach_realized` (`FrameworkUniversalReach.lean` line 153).

Bundle-field discipline (quoted from the Lean record’s field declarations, lines 99–137): of the 17 fields in `FrameworkUniversalReach`, **15 are typed as `: True`** (proveness tags carrying no mathematical content at the record level). The two fields that *do* carry non-trivial framework Props at the record level are `brocard_attack_realized` and `hadwiger_nelson_attack_realized`. The substantive per-axis content for the other 14 non-Clay problems and the seven Clay axes lives in separate framework-attack modules cited by the record file header. The 23-problem record is a single citation point indexing these modules; reading the record itself as a Clay-Standard discharge is incorrect.

7 Independent verification posture

The Lean 4 proof objects for the theorems cited in this paper type-check in the Lean 4 kernel with kernel axioms `[propext, Classical.choice, Quot.sound]` and zero project axioms. The `PF_Lean4Lean` harness provides **build-hash robustness across two Lake packages**: the canonical theorems are re-bound in a separate package and rebuilt from source. **It is not an independent type-checker written in another language.** Quoted verbatim from `PF_Lean4Lean/PF_L4L/Referee/ClayVerificationHarness.lean` lines 47–56:

“This is the same Path-C protocol as `FlagshipReverification` and `V2AndMasterReverification`: a second pass through Lean’s kernel from a separate package with an independent build hash. It is NOT a separate type-checker written in another language. The honest scope of each individual closure is the one fixed by that closure’s own `*_honestScope` / `*_honest_scope` marker in the canonical PF library... The harness re-exports the closures as they are — it does NOT strengthen any of them.”

A Coq mirror provides cross-prover structural parity; approximately 75% of definitions in the Coq mirror are typed `: Prop := True`, i.e. the Coq tree is **parity-only** documenting that the naming skeleton admits cross-prover encoding, not independent semantic verification of the Lean content.

8 Honest scope

Per the file-level honest-scope marker (`(sec : substrate – honest – scope)`), PF is a *substrate-level monograph in Grothendieck mode*.

Remark 8.1 (Honest scope on the RH axis, foregrounded). This paper, and the framework it foregrounds on the RH axis, are *not* a literal mathlib-tier discharge of the Riemann Hypothesis in the Clay form. The substrate-level RH content is conditional on the `HilbertPolyaProgramConjecture` (preserved openly in the Hilbert–Pólya identification module) and on the Mayer 1991 §3 spectral correspondence (`Mayer1991_SymmetricQuotientHasZetaSpectrum`, preserved openly in the Mayer transfer-operator formalization module). The five-formulation Hilbert–Pólya collapse (theorem 3.1) is unconditional at the substrate level; the collapse target is the published HP program. The substrate-level Ch 17 concrete witness exhibits a 3×3 self-adjoint Hamiltonian with spectrum $\{1, 2, 3\}$; the literal continuum Hilbert–Pólya operator on an infinite-dimensional space remains the named open content. The 23-problem reach record is, at the record level, 15 fields typed `: True` plus 2 fields carrying non-trivial Props (Brocard, Hadwiger–Nelson); the substantive per-axis content lives in the per-axis attack modules cited by the record.

Remark 8.2 (Cross-Millennium α -skeleton: framework-internal). The eleven cross-Millennium arithmetic compatibility identities that pin $\alpha_{\text{RH}} = 3/2$ are framework-internal: they are algebraic facts about the nine *chosen* α -values, each provable by `ring/norm_num` on the definitions. The framework’s interpretation that each α is the canonical eigenvalue parameter for its Millennium operator is, per the file header of `PF.CrossMillenniumSharedInvariants`, “conjectural and lives elsewhere.”

Remark 8.3 (Falsifiability). The framework commits to eight typed falsifiability conditions in `PF.Referee.FrameworkFalsifiabilityConditions`. As of the 2026-06-04 referee-proof literature audit: **zero of eight triggered; two actively supported by 2024–2026 literature** (F_3 dark-energy direction and $F_6 \Omega_\Lambda \in [0.65, 0.75]$).

9 Conclusion

This Phase 1 paper presents a substrate-level reformulation of the Riemann Hypothesis on the framework's T_3^{sym} encoding, machine-verified in Lean 4 with kernel-only axioms. The substrate-level RH bridge is conditional on two named typed residuals; it is not a literal mathlib-tier discharge of RH at the Clay form. The RH axis is the natural *entry vector* to the framework because of the pre-built GRH-conditional scaffold in analytic number theory; the framework's contribution is structural reformulation on a substrate-level encoding, not prize-eligible discharge.

The same substrate carries substrate-level encodings of the other five remaining Clay axes; the simultaneous-closure mechanism on canonical encodings is the subject of the master paper [1]. The framework's reach beyond the Clay set (23 named open problems via a single-citation record), the empirical anchors (143-problem coherence and IBM Quantum 9-way), and the Phase 4 substrate-level content on consciousness, ZPE, and cosmology are presented in the companion papers. **Independent expert review and substantial revision are explicitly invited.**

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